DocLine WhatsApp — Complete Integration Guide

Product: DocLine (GlucoGuide) — Healthcare clinic appointment booking over WhatsApp
Version: 1.0 · August 2026
Audience: WhatsApp BSP / Meta API provider, DocLine ops, clinic onboarding, engineering
Related docs: `WhatsApp-How-It-Works.md` , `Whatsapp-Meta-Rules.md` , `decisions.md`

Document control

Field	Value
Purpose	End-to-end WhatsApp setup: pool numbers, clinic-owned numbers, provider workflow, scaling, compliance
Integration	Meta WhatsApp Cloud API (direct or via BSP)
Webhook	Single URL for all clinics: `/api/v1/procto/whatsapp/webhook`
Template category	UTILITY only
Meta approval SLA	Typically 24–72 hours (display name + templates)

Table of contents

1. Executive summary
2. How DocLine WhatsApp works
3. Three ways to set up a clinic number
4. Scaling beyond 20 numbers
5. Managing N WABA IDs
6. Meta Business account and payment
7. Clinic-owned number workflow
8. Restrictions when linking to your Meta app
9. Approval timelines
10. Provider setup workflow
11. Webhook configuration
12. Message templates (all 10)
13. Meta compliance rules
14. Environment variables and credentials
15. Go-live checklists
16. Testing checklist
17. Architecture and code reference
18. FAQ
19. Provider deliverables sign-off

---

1. Executive summary

DocLine connects clinic WhatsApp numbers to an automated booking bot on our server. **The bot does not run inside the doctor's WhatsApp app.**

```
Patient → Clinic WhatsApp number → Meta Cloud API → DocLine server → Bot reply
```

Key facts:

- One WhatsApp number = **one clinic (practice)** — all doctors in that clinic share it
- Doctors do **not** scan a QR to activate the bot
- QR codes (`wa.me/91XXXXXXXXXX`) are for **patients** to open chat faster
- Proactive messages (confirmations, reminders) require **approved UTILITY templates**
- Session replies (when patient messages first) work free-form within Meta's 24-hour window

Provider responsibilities: Meta app, WABA, webhook, tokens, templates, number registration
DocLine responsibilities: Bot logic, routing, admin assign, clinic Billing UI
Clinic responsibilities: Subscribe to plan, connect own number (or use assigned pool number)

Typical full go-live: Connect can be same-day; Meta display name + template approval usually **1–3 business days**.

---

2. How DocLine WhatsApp works

2.1 The one thing to remember

The bot lives on **our server**. When a patient sends a message, Meta forwards it to us, and we reply automatically. Once the clinic number is **connected and live**, the bot answers patients automatically. No doctor scan needed.

2.2 What you wanted vs what actually works

Idea	Works?	Why
Patients message clinic number → bot books for them	**Yes**	This is what we built
Doctor scans QR to "turn on" the bot	**No**	Meta does not work that way
QR on clinic poster so patients open chat	**Yes**	Optional marketing only

QR is for patients to open chat faster. It does not connect or activate the bot.

2.3 When the bot is active

All of the following must be true:

1. Clinic has **Starter+ plan** (WhatsApp feature enabled)
2. A **clinic WhatsApp number** is linked to the practice
3. Meta has **approved** the clinic name and message templates (for full proactive alerts)
4. Our **webhook** is connected to Meta

2.4 Inbound message flow

```
Patient → Clinic WA # → Meta Cloud API → DocLine webhook → Bot orchestrator → Reply
```

DocLine routes inbound messages by **Phone Number ID** (globally unique), not by WABA ID.

2.5 Outbound message flow

```
DocLine booking event → Notification worker → Meta template API → Patient WhatsApp
```

Outside the 24-hour customer service window, **only approved templates** may be sent.

2.6 What the bot does for patients

When the line is live, patients can message and get:

- Register
- Book appointment
- Check status
- Cancel / reschedule
- Voice notes (speech-to-text)
- Reminders (via approved templates)
- **STOP** to opt out of alerts

Doctors do not use the bot chat. If a doctor messages the clinic line, they get a message to use the web portal (`/doctor/queue` , `/doctor/practice`).

2.7 Customer service window vs templates

Situation	Allowed
User messaged the business line within **24 hours**	Free-form session replies (booking bot)
Outside 24h, or first outbound reminder	**Template** only
User replies after a template	New 24h window opens

---

3. Three ways to set up a clinic number

Option 1 — Pool number (easiest for pilot)

We give the clinic a number from our pool.

- Doctor shares that number with patients (card, poster, profile)
- Doctor keeps their personal WhatsApp separate
- Ops assigns the number in admin
- **Best for:** First 3–5 doctor pilot
- **Counts against 20-number Meta cap:** Yes

```
Subscribe → Ops assigns number → Go live → Patients message that number → Bot works
```

Option 2 — Clinic's own number (recommended for scale)

Same number — bot plus doctor can both use WhatsApp (coexistence).

Doctor must use the **WhatsApp Business app** (not personal WhatsApp).

Flow (self-serve in Billing):

1. Doctor subscribes to **Starter+** under Practice → **Billing**
2. Enters **clinic WhatsApp number**
3. Clicks **Connect with Meta** (Embedded Signup) or enters **Meta Phone Number ID**

4. Meta webhook must point to DocLine server
5. UTILITY templates must be **APPROVED** on that clinic's WABA (24–72h)

Doctor can still reply manually in the WhatsApp Business app. **Bot handles** booking, status, cancel, reminders.

Meta calls this **Coexistence**.
Official doc: https://developers.facebook.com/docs/whatsapp/embedded-signup/custom-flows/onboarding-business-app-users/

Does not count against pool cap when properly onboarded per clinic WABA.

Option 3 — Personal WhatsApp on clinic number (hard — avoid)

- Doctor must **delete** personal WhatsApp on that number
- Number is moved to Meta Business API
- Personal WhatsApp cannot stay on the same number
- Old chat history is not kept on the bot side

Better path: Move to WhatsApp Business app first, then use Option 2.

Quick decision guide

Ask the doctor: *Is your clinic number on personal WhatsApp or WhatsApp Business app?*

Answer	Recommendation
Personal WhatsApp	Pool number (Option 1) OR migrate to Business app first
WhatsApp Business app	Option 2 — Connect with Meta
Fastest pilot	Option 1 — pool number

---

4. Scaling beyond 20 numbers

Meta limits registered phone numbers per business portfolio:

Stage	Typical cap
New / unverified	**2 numbers**
Business verified	**20 numbers**
Beyond 20	Request increase via Meta Business Support

Scaling strategy:

1. **Pilot:** Pool numbers (Option 1) — up to ~20 clinics on one verified portfolio
2. **Growth:** Clinic-owned numbers (Option 2) — each clinic brings their own number
3. **If more central numbers needed:** Multiple WABAs + Meta support ticket for limit increase
4. **Optional:** BSP partner (Gupshup, Interakt, WATI, AiSensy)

Do not scale by: stuffing unlimited clinics into one Meta account's 20-number pool.

One number can handle high message volume via Meta **messaging tiers** (daily conversation limits). The 20 cap is about **distinct clinic lines**, not messages per line.

---

5. Managing N WABA IDs

When using clinic-owned numbers at scale, expect **one WABA per clinic** (not one WABA for all clinics).

```
Clinic A → WABA_A → Phone Number ID_A
Clinic B → WABA_B → Phone Number ID_B
Clinic C → WABA_C → Phone Number ID_C
```

What works today without code changes

Concern	Status
Inbound routing	**Works** — by Phone Number ID from webhook
Outbound sends	**Works** — platform token + per-clinic Phone Number ID
One webhook for all	**Works**
WABA ID stored in DB	**Gap** — use spreadsheet; add `wabaId` column later
Templates per WABA	**Main ops work** — 10 templates must be approved on each WABA

Three-layer management model

Layer 1 — Provider (Meta/BSP):

- One Meta App + one webhook URL
- One system/partner token with access to all onboarded WABAs
- Onboards each clinic → creates WABA + Phone Number ID
- Auto-syncs 10 UTILITY templates on each new WABA (ask provider to confirm)

Layer 2 — DocLine registry (ops):

practice_name	clinic_phone	waba_id	phone_number_id	display_name_status	templ

Layer 3 — DocLine runtime:

- Webhook routes by `phone_number_id`
- Send uses token + clinic's `phone_number_id`
- Same bot logic for all clinics

Questions to ask your provider

1. Does **one access token** work for all clinic WABAs?
2. Is **one webhook** sufficient for all WABAs?
3. Do you **auto-submit 10 UTILITY templates** on each new clinic WABA?
4. Does Embedded Signup return `waba_id` + `phone_number_id` ?
5. Is **coexistence** enabled for WhatsApp Business app users?

Avoid: per-clinic separate API tokens (not supported in DocLine today).

---

6. Meta Business account and payment

When **you** create Meta Business and add payment:

```
YOUR Meta Business Portfolio (verified, payment added)
    │
    ├── Your Meta Developer App (WhatsApp product)
    │       └── Webhook → DocLine server
    │       └── System user token → DocLine backend
    │
    └── Clinic onboarding (Embedded Signup / provider)
            ├── Clinic A → WABA_A → Phone + Phone Number ID_A
            ├── Clinic B → WABA_B → Phone + Phone Number ID_B
            └── Clinic C → WABA_C → Phone + Phone Number ID_C
```

You pay Meta for WhatsApp conversation charges. **Clinics pay DocLine** subscription (Starter ₹1499+, Growth ₹2999, Clinic ₹4999).

Clinics do **not** need to:

- Create their own Meta Business account
- Add their own card to Meta
- Manage WhatsApp API billing

Who manages what

Task	You / provider	Clinic
Meta Business account & payment	Yes	No
Meta App, webhook, token	Yes	No
Connect their WhatsApp number	Assist	Clicks "Connect with Meta"
Share number with patients	—	Cards, poster, profile
Reply manually to patients	—	WhatsApp Business app
Booking bot, reminders	DocLine	Automatic
Display name + templates	Provider submits	Provides clinic name

---

7. Clinic-owned number workflow

Provider one-time setup

- Meta Business verified
- Payment method added
- Meta App + WhatsApp API enabled
- Webhook → DocLine production URL
- Embedded Signup with `whatsapp_business_app_onboarding` (coexistence)
- 10 UTILITY templates ready for sync to each new WABA

Per-clinic connect (doctor steps)

1. Subscribe to DocLine **Starter+**
2. Ensure clinic number is on **WhatsApp Business app**
3. Open **Practice → Billing → WhatsApp booking line**
4. Enter 10-digit clinic mobile

5. Click **Connect with Meta**
6. Complete Meta popup (Facebook login + authorize)
7. Status shows **Live** — copy `wa.me` link for patients

After connect

- **Patients** message clinic number → DocLine bot replies
- **Doctor** can use WhatsApp Business app on same number (coexistence)
- **Provider** ensures templates approved on that clinic's WABA
- **Ops** updates registry when Meta approvals complete

---

8. Restrictions when linking to your Meta app

Number eligibility

Restriction	Detail
Personal WhatsApp	Cannot use coexistence — need Business app
Number on another API/BSP	Must migrate/unlink first
One API per number	Cannot be on two providers simultaneously
Must own number	Real SIM; SMS/voice OTP capable
Business app version	2.24.17+ for coexistence

Partner / lock-in

- Number/WABA is tied to **your Meta app + provider**
- Switching BSP later requires Meta number migration
- Clinic authorizes via Facebook login during Embedded Signup

Coexistence limitations (Business app + API)

Feature	After linking
Throughput	~20 messages/second cap when app + API share number
Template messages	Sent **only via API** (DocLine)
Broadcast lists in app	Become read-only
Disappearing / view-once messages	Disabled in 1:1 chats
Live location	Disabled in 1:1 chats
Edit / delete messages	Disabled in 1:1 threads
Linked devices	Unlinked during onboarding; re-link up to 4 (Windows/WearOS limited)
App must stay active	Open Business app every ~13–14 days

Messaging policy

- 24-hour window for free-form replies
- UTILITY templates only for appointment alerts (no MARKETING miscategorization)
- Opt-in before first proactive template
- STOP must work
- No medical advice on bot
- Display name must match real clinic
- Quality rating — Low quality can pause the number

DocLine platform limits

- One number per clinic (practice)
- Changing number requires support
- Starter+ subscription required

---

9. Approval timelines

Step	Who	Typical time
Connect with Meta in DocLine	Automatic	**Minutes**
Phone OTP verification	Meta	**Minutes**
DocLine shows **Live**	DocLine (on connect)	**Instant**
Display name approval	Meta review	**24–72 hours**
Template approval (per WABA)	Meta review	**24–72 hours**

What works before full Meta approval

Feature	Day 0 (connect)	After templates approved
Patient sends HELP/BOOK	Yes	Yes
Session bot (24h window)	Yes	Yes
Web booking → WhatsApp template confirm	Maybe fallback	Yes
Automated reminders outside 24h	No / fallback	Yes

Messaging for clinics

After connect:
"Your number is connected. Patients can message and the booking bot will reply. Meta is reviewing your clinic name and message templates — usually 1–3 business days."

After full approval:
"WhatsApp booking is fully live — bot + confirmations + reminders."

Never promise: instant full go-live including all automated reminders.

---

10. Provider setup workflow

Phase 1 — Meta platform setup

1. Complete **Business Verification** in Meta Business Suite
2. Create Meta Developer App (Business type) + add WhatsApp product
3. Create WABA; record WABA ID
4. Create System User; generate permanent token with `whatsapp_business_management` + `whatsapp_business_messaging`
5. Configure Embedded Signup with `whatsapp_business_app_onboarding`

Phase 2 — Phone numbers

Pool (Model A): Register numbers on platform WABA; record Phone Number ID per number.

Clinic-owned (Model B): Clinic connects via Embedded Signup; provider verifies number appears in WhatsApp Manager.

Phase 3 — Webhook

See Section 11.

Phase 4 — Templates

See Section 12. Submit all 10 on each clinic WABA.

Phase 5 — Credentials handoff

See Section 14.

---

11. Webhook configuration

Field	Value
Callback URL	`{API_BASE}/api/v1/procto/whatsapp/webhook`
Verify token	Agreed string — DocLine sets `WHATSAPP_VERIFY_TOKEN`
Signature	`X-Hub-Signature-256` validated with `WHATSAPP_APP_SECRET`

Subscribe to: `messages`, `message_template_status_update`, `account_update`, `phone_number_quality_update`

One webhook for all clinics. DocLine routes by `metadata.phone_number_id` in payload.

Test: Patient sends HELP → bot reply within 30 seconds.

---

12. Message templates (all 10)

All **UTILITY**, language `en`, **named parameters**. Must be approved on **each clinic WABA**.

Internal key	Meta name	Purpose
BOOKING_CONFIRMATION	docline_booking_confirm	After book
REMINDER	docline_appointment_reminder	Before slot

CANCELLATION	docline_booking_cancelled	Cancel
RESCHEDULE	docline_booking_rescheduled	Time change
OVERRIDE_IMPACT	docline_schedule_change	Schedule block impact
QUEUE_ALERT	docline_queue_ready	You're up next
CLINIC_NEW_BOOKING	docline_clinic_new_booking	Staff alert
CLINIC_CANCEL	docline_clinic_booking_cancelled	Staff alert
CLINIC_RESCHEDULE	docline_clinic_booking_rescheduled	Staff alert
CLINIC_QUEUE_ALERT	docline_clinic_queue_alert	Staff alert

Template 1: docline_booking_confirm

```
Your appointment at {{practice_name}} is confirmed.
When: {{when_label}}
Doctor: {{provider_name}}
Location: {{location}}
Reply CANCEL to cancel, RESCHEDULE to change, or STATUS to view.
Reply STOP to opt out of appointment alerts.
```

Template 2: docline_appointment_reminder

```
Reminder: you have an appointment at {{practice_name}}.
When: {{when_label}}
Doctor: {{provider_name}}
Please arrive 10 minutes early.
Reply STOP to opt out of appointment alerts.
```

Template 3: docline_booking_cancelled

```
Your appointment at {{practice_name}} ({{when_label}}) has been cancelled.
Reply BOOK to schedule again, or STOP to opt out of appointment alerts.
```

Template 4: docline_booking_rescheduled

```
Your appointment at {{practice_name}} was rescheduled.
Previous: {{old_when}}
New: {{when_label}}
Doctor: {{provider_name}}
Reply STOP to opt out of appointment alerts.
```

Template 5: docline_schedule_change

```
Schedule update from {{practice_name}}.
Your visit ({{when_label}}) is affected{{reason_suffix}}.
```

```
Reply RESCHEDULE to pick a new time, or CANCEL to cancel.
Reply STOP to opt out of appointment alerts.
```

Template 6: docline_queue_ready

```
You are up next at {{practice_name}}{{token_suffix}}.
Please be ready at the clinic.
Reply STOP to opt out of appointment alerts.
```

Template 7: docline_clinic_new_booking

```
New booking at {{practice_name}}.
Patient: {{patient_name}}
When: {{when_label}}
Doctor: {{provider_name}}
Channel: {{channel}}
```

Template 8: docline_clinic_booking_cancelled

```
Booking cancelled at {{practice_name}}.
Patient: {{patient_name}}
Was: {{when_label}}
```

Template 9: docline_clinic_booking_rescheduled

```
Booking rescheduled at {{practice_name}}.
Patient: {{patient_name}}
Previous: {{old_when}}
New: {{when_label}}
```

Template 10: docline_clinic_queue_alert

```
Patient {{patient_name}} is now in progress at {{practice_name}} ({{when_label}}).
```

Template rules

- Category: UTILITY only — never MARKETING
- Include STOP opt-out on patient templates
- No promos, discounts, upsells
- Full bodies in `backend/src/services/procto/whatsapp.metaTemplates.ts`

---

13. Meta compliance rules

Absolute rules (ban risks)

1. Never send free-form proactive text outside 24h window — use approved templates
2. Never miscategorize MARKETING as UTILITY
3. Opt-in before first proactive template
4. Honour STOP (sets `waOptOut`; blocks proactive alerts)

5. No medical advice on bot — escalate to HUMAN
6. Disclose AI on first message
7. Never send passwords, OTPs, or payment card data on WhatsApp
8. One quality WABA per brand — don't spam same user from multiple numbers
9. Display name must match real clinic
10. No cold outreach to non-opted-in contacts

Bot session behaviour

Do	Don't
AI disclosure on first message	Pretend to be a human doctor
FAQ from clinic faqKb only	Invent clinical advice
HUMAN handoff	Diagnose / prescribe
HELP menu for bookings	Blast marketing
STOP / START	Ignore opt-out

Incident response

1. Stop proactive sends for affected WABA
2. Read Meta `account_update` webhook / email
3. Pause offending templates; fix copy; resubmit
4. Audit opt-out misses
5. Resume only after quality recovers

---

14. Environment variables and credentials

```
WHATSAPP_ACCESS_TOKEN=<permanent system user token>
WHATSAPP_WABA_ID=<platform waba id — for pool lookup>
WHATSAPP_PHONE_NUMBER_ID=<default pool number id, if any>
WHATSAPP_VERIFY_TOKEN=<agreed verify string>
WHATSAPP_APP_SECRET=<app secret>
WHATSAPP_API_VERSION=v21.0

META_APP_ID=<app id>
META_APP_SECRET=<app secret>
META_EMBEDDED_SIGNUP_CONFIG_ID=<config id>
META_EMBEDDED_SIGNUP_FEATURE=whatsapp_business_app_onboarding
```

Send credentials via secure channel only.

---

15. Go-live checklists

Platform (provider — one time)

- [ ] Meta Business verified
- [ ] App + WABA + System User token

- [ ] Embedded Signup with coexistence
- [ ] Webhook verified on production
- [ ] Credentials sent to DocLine

Per clinic

- [ ] Starter+ subscription active
- [ ] Phone registered on Meta WABA
- [ ] Connect with Meta completed (or pool number assigned)
- [ ] Phone Number ID stored in DocLine
- [ ] Display name APPROVED in WhatsApp Manager
- [ ] 10/10 templates APPROVED on clinic WABA
- [ ] Test: HELP → bot reply
- [ ] Test: BOOK → booking flow
- [ ] Test: web book → template confirmation (after templates approved)
- [ ] Test: STOP opts out
- [ ] wa.me link shared with clinic

---

16. Testing checklist

#	Test	Pass criteria
1	Webhook GET verify	Meta shows verified
2	Inbound HELP	Bot reply < 30s
3	Inbound BOOK	Booking flow starts
4	Voice note	Transcribed + processed
5	Template confirm send	Patient receives formatted message
6	STOP	No further proactive alerts
7	Multi-clinic routing	Each Phone Number ID routes to correct practice
8	Invalid webhook signature	401 rejected

---

17. Architecture and code reference

Architecture decisions

Decision	Choice
WABA model (pilot)	One platform WABA, one phone per practice
WABA model (scale)	N WABAs — one per clinic via Embedded Signup
Provisioning SLA	24–72h (Meta-controlled)
Routing key	Phone Number ID (not WABA ID)

Webhook	Single URL for all clinics

Code files

What	File
Assign numbers	`backend/src/services/procto/numberPool.service.ts`
Connect own number	`numberPool.service.ts` → `connectOwnNumber`
Embedded Signup	`backend/src/services/procto/metaWhatsApp.service.ts`
Incoming messages	`backend/src/services/procto/whatsapp.service.ts`
Booking conversation	`backend/src/services/procto/whatsapp.orchestrator.ts`
Template catalog	`backend/src/services/procto/whatsapp.metaTemplates.ts`
Send API	`backend/src/services/procto/whatsapp.client.ts`
Webhook routes	`backend/src/routes/procto/whatsapp.ts`
Billing / connect UI	`frontend/components/ui/procto/PracticeWhatsAppConnect.tsx`

API endpoints

Endpoint	Method	Purpose
`/api/v1/procto/whatsapp/webhook`	GET/POST	Meta webhook
`/api/v1/procto/practices/:id/whatsapp/connect`	POST	Clinic connects own number
`/api/v1/procto/practices/:id/whatsapp/resolve`	POST	Auto-resolve Phone Number ID
`/api/v1/procto/admin/numbers`	GET/POST	Pool management
`/api/v1/procto/admin/numbers/assign`	POST	Assign pool number
`/api/v1/procto/admin/templates/sync-meta`	POST	Sync template catalog

Plans

- **Starter** ₹1499/mo — WhatsApp line, directory, 150 bookings/mo
- **Growth** ₹2999/mo — + reminders, reviews, multi-doctor
- **Clinic** ₹4999/mo — + queue analytics, dedicated support

---

18. FAQ

Does the doctor scan a QR to turn on the bot?
No. QR is for patients only.

One number per doctor or per clinic?
One number per **clinic (practice)**.

Can personal WhatsApp be used?
No for coexistence. Use Business app or pool number.

One webhook for all clinics?
Yes. DocLine routes by Phone Number ID.

How do reminders work?
DocLine calls `POST /v21.0/{phone_number_id}/messages` with approved templates.

What if we hit 20 pool numbers?
Use clinic own number (Option 2), request Meta limit increase, or add WABAs.

Who pays Meta?
Platform (DocLine). Clinics pay DocLine subscription.

Is there an approval wait?
Connect is minutes. Display name + templates: typically 1–3 days.

N WABA IDs — is that normal?
Yes at scale. One WABA per clinic. Route by Phone Number ID.

---

19. Provider deliverables sign-off

#	Deliverable	Provider	DocLine	Date
1	WABA ID(s)	☐	☐	
2	App ID + App Secret	☐	☐	
3	Permanent access token	☐	☐	
4	Embedded Signup config ID	☐	☐	
5	Webhook verified	☐	☐	
6	Phone Number IDs + mobiles registry	☐	☐	
7	Display names approved	☐	☐	
8	All 10 templates APPROVED per WABA	☐	☐	
9	HELP test passed	☐	☐	
10	Coexistence enabled	☐	☐	

Provider signature: _____ Date: _____
DocLine signature: _____ Date: _____

What the provider must NOT do

Do not	Reason
MARKETING templates as UTILITY	Ban / recategorization risk
Cold outreach	Policy violation

Passwords/OTP on WhatsApp	DocLine forbidden
Separate webhook per clinic	Single webhook architecture
Promise instant full go-live	Meta approval 1–3 days
Personal WhatsApp coexistence	Requires Business app

End of document

DocLine / GlucoGuide · WhatsApp Complete Integration Guide v1.0 · August 2026

Sources: WhatsApp-How-It-Works.md, Whatsapp-Meta-Rules.md, decisions.md, and DocLine backend implementation.